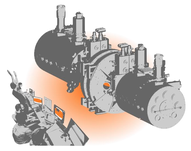
Beam-Divergence Spectroscopy on the WHAM Neutral Beam Injector using a Voltage-History Deconvolution Framework

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Background & Motivation

Original analysis

- Doppler shift spectroscopy (DSS) was only used to measure neutral beam injection (NBI) species composition

D^+ (f1): 70.6%, D_2^+ (f2): 23.1%, D_3^+ (f3): 6.4%

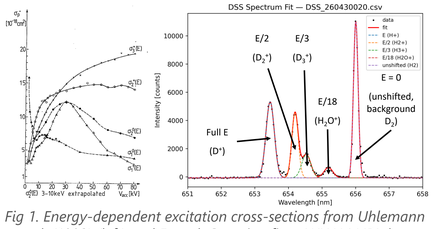


Fig 1. Energy-dependent excitation cross-sections from Uhlemann et al. (1993) (left) and 5-peak Gaussian fit to WHAM NBI data.

- Array of secondary electron emission (SEE) detectors was used at the beam dump to measure the spatial distribution of beam neutrals.

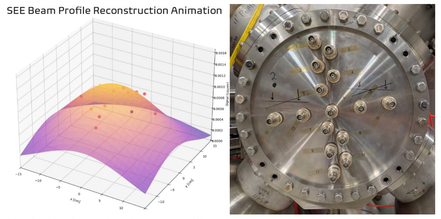


Fig 2. 2D Gaussian beam profile reconstruction from SEE detectors (left) and WHAM NBI SEE detector array (right).

Beam divergence from Doppler-shift spectroscopy (DSS)

- The width of the Doppler-shifted D_α line shape encodes beam divergence $\boldsymbol{\varepsilon/\sigma_{x'}}$.
- Standard analysis (Deka et al. 2018) fits a single Gaussian peak per species, then removes known broadenings in quadrature, including a scalar voltage-ripple term:

$$\Delta\lambda_W^2 = \Delta\lambda_N^2 + \Delta\lambda_I^2 + \Delta\lambda_f^2 + \Delta\lambda_\varepsilon^2 + \Delta\lambda_{\text{ripple}}^2$$

$$\Delta\lambda_{\text{ripple}} = \frac{e\lambda_0}{\beta mc^2}(\beta - \cos\alpha)\delta V$$

$$\varepsilon = \frac{\sqrt{\Delta\lambda_W^2 - \left(\Delta\lambda_N^2 + \Delta\lambda_I^2 + \Delta\lambda_{\text{coll}}^2 + \Delta\lambda_f^2 + \Delta\lambda_{\text{ripple}}^2\right)}}{\lambda_0\beta\sin\alpha/\sqrt{1-\beta^2}}$$

WHAM's HVPS is not a flat plateau

- The ripple term assumes a stable plateau with a small, symmetric ripple (ROBIN source: 14–25 kV, ~1 %)
- Measured V(t) on the WHAM NBI stand ramps, decays, and carries a sawtooth; flat-top mean 19.93 kV, 1.12 kV spread (5.6 %) over the beam-on gate
- Wavelength smear: σ_K = 0.199 nm ≈ 3× the instrumental width, skew −5.93; too large and too asymmetric for a scalar symmetric term

⇒ The quadrature ripple correction must be replaced, not rescaled.

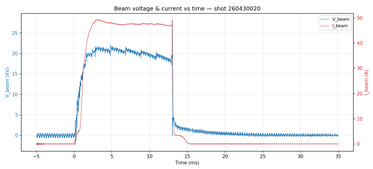


Fig 3. WHAM HVPS voltage & current trace; not the plateau & ripple Deka's model assumes.

Analysis Upgrades

Line shape modeled as a convolution

- Each instant emits an intrinsic Gaussian *g* (every broadening term except voltage), centered at the instantaneous Doppler shift Δλ(V(t))
- Written as a mathematical convolution:

$$\mathbf{I}_{\text{obs}}(\boldsymbol{\lambda}) = (\mathbf{K} \circledast \mathbf{g})(\boldsymbol{\lambda})$$

- **K** = emission-weighted histogram of instantaneous Doppler shifts, built once per shot from measured V(t)
- Final observed line shape $\mathbf{I}_{\text{obs}}(\boldsymbol{\lambda})$ is the sum of the Gaussian peaks at each voltage weighted by current, i.e., a higher current (more particles traveling down the beamline) will emit more doppler-shifted light, causing a larger amplitude Gaussian to be measured at that voltage

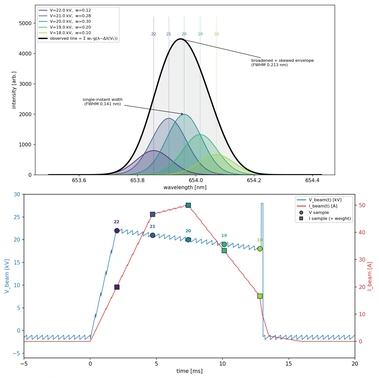


Fig 4. Current-weighted voltage convolution demonstration.

Forward-fit kernel

- Forward-fit: K is convolved into a known-shape g and compared to the data, so the fitted width (s ≡ σ_{true}) is the intrinsic width
- Fourier deconvolution $\hat{g} = \hat{I}_{\text{obs}}/\hat{K}$ is avoided due to numerical instability when $\hat{K}$ moves through 0

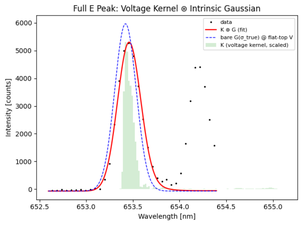


Fig 5. Full-energy peak forward-convolution fit to data (red) vs deconvolved bare Gaussian (blue). Blue curve indicates what the expected line shape of full energy particles would be without voltage droop.

DSS & SEE cross-validation

- SEE → spatial divergence: $\sigma_x(\mathrm{L}) = \sqrt{2.27\mathrm{e}{-04} + 5.64\sigma_{x'}^2}$
- **Beam envelope equation:**

$$\sigma_x(L) = \sqrt{\sigma_{x(0)}^2 + 2L\sigma_{xx'} + L^2\sigma_{x'}^2} = \sqrt{\sigma_{x(0)}^2\left(1 - \frac{L}{F}\right)^2 + L^2\sigma_{x'}^2}$$

- Yields combined effect of distance L, beam focal length F, $\sigma_{xx'}$, and $\sigma_{x'}$
- $\sigma_{x(0)} \approx \frac{R}{2}, \qquad \sigma_{xx'} \approx -\frac{\sigma_{x(0)}^2}{F}$

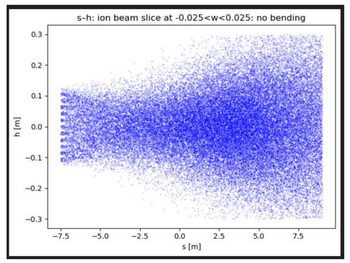


Fig 6. NBI particle simulation; uniform particle distribution at the extraction plane, Gaussian particle distribution at the beam dump (SEE detector measurement)

Campaign Design

1. Design an experimental campaign that measures DSS divergence across a diverse set of voltages (*V*) and perveances (Π), the two main control levers of beam divergence
2. Fit these data to a semi-empirical model $\sigma_{x'}(V, \Pi)$ that outputs beam divergence $\sigma_{x'}$ as a function of voltage and perveance
3. Plug time-resolved voltage and current from WHAM HVPS into the fitted model to calculate time-resolved beam divergence
4. Use the beam envelope equation to cross-validate DSS & SEE measurements

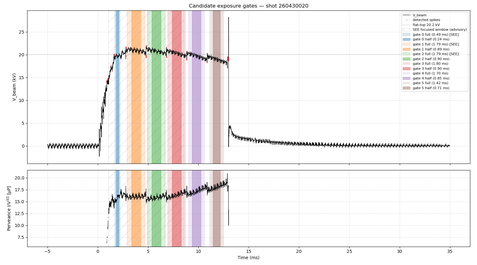
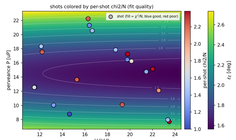


Fig 7. Measured V(t) (shot 260430020) decaying ramp sawtooth with DSS time gates.

Campaign points

- Voltage scan: V_PG = 14 / 17 / 20 / 25 kV
- Perveance scan: arc = 4.7 / 4.1 / 3.5 / 2.9



Gate	Window [ms]	Width [ms]
g1	3.15 – 4.65	1.50
g3	7.17-8.67	1.50
g5	11.30-12.80	1.50

Fig 8. Campaign points taken in voltage, perveance space (left) and table of time gates used throughout the course of one shot (right).

Model fit

- Model: $\sigma_{x'}^2(V, \Pi, arc) = \underbrace{a + b(\Pi - \Pi_0)^2}_{\text{optics}} + \underbrace{\frac{T_\perp[arc]}{2f_S V}}_{\text{thermal}}$
- Fit: $\epsilon_s^2 = 6.60\mathrm{e}{-04} + 1.86\mathrm{e}{-04}(\Pi - 14.5)^2 + \frac{T_\perp[arc]}{2f_S V}$
- $\alpha = 32.86°$ (E-constrained), Floor divergence $\sqrt{a} = 1.47°$
- Global $\chi^2/N = \frac{1}{N}\sum_i\left[\frac{\text{data}_i - \text{model}_i}{\sigma_i}\right]^2 = 1.77$

Fisher Information

- $\mathcal{F}(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right]$
- $X \equiv$ observed data
- $\theta \equiv$ parameter; true state of nature
- Exact kernel **K** removes bias ($\sigma_{x'}$ not systematically inflated), but does not remove variance → wide voltage kernel is unbiased but imprecise
- $\delta\sigma_0 = \delta\sigma_{obs}\sqrt{1 + \frac{\sigma_K{}^2}{\sigma_0}}$

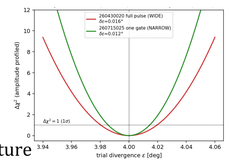


Fig 9. Likelihood well for a wide (red) and narrow (green) voltage kernel, indicating greater ε uncertainty with a wider voltage kernel

Results

Fitted Model

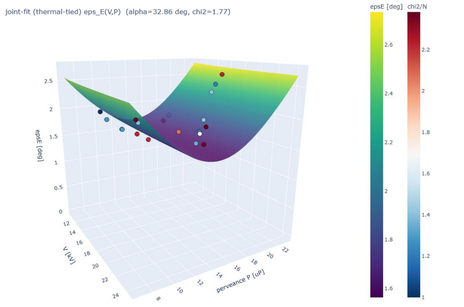


Fig 10. Global model fitted surface. Beam divergence as a function of voltage and perveance.

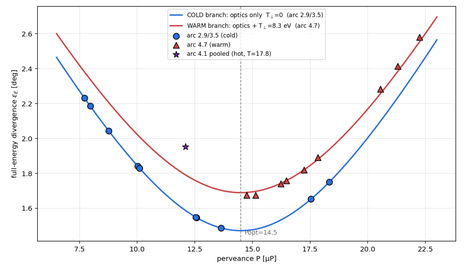


Fig 11. Campaign divergence as a function of beam perveance located on either high or low arc curves.

Voltage kernel adds significant line width

- 2.83x greater for full pulse vs gated shot

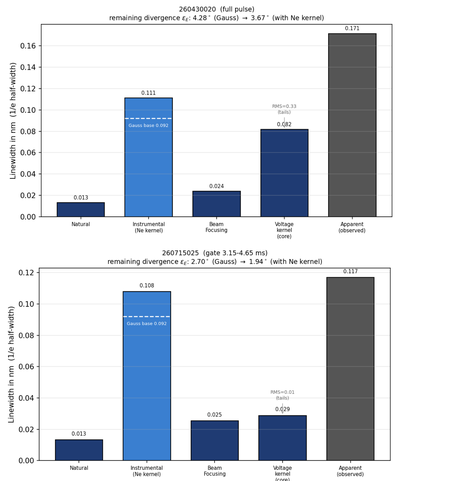


Fig 12. DSS broadening terms of full shot (left) vs gated shot (right).

Conclusions

- Reframed DSS divergence as a forward convolution with a voltage-history kernel; voltage smear removed by construction
  - Convolution model reduces to Deka model in the plateau / narrow-ripple limit
- Showed strong agreement between time-resolved SEE and DSS beam divergence

DSS / SEE cross-validation

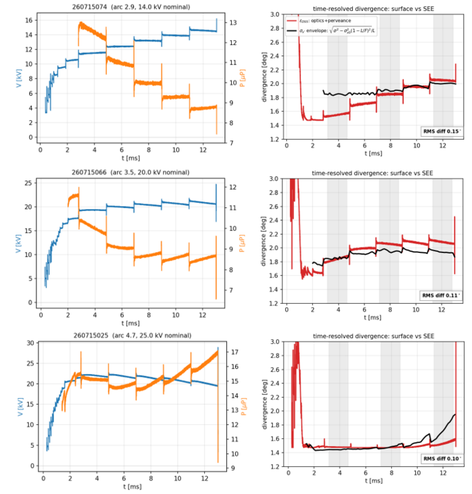


Fig 13. Voltage & perveance traces (column 1) for three shots in different operational regimes (rows 1, 2, 3) and cross-validated time-resolved beam size σ_x from DSS & SEE (column 2).

Time-resolved species fractions

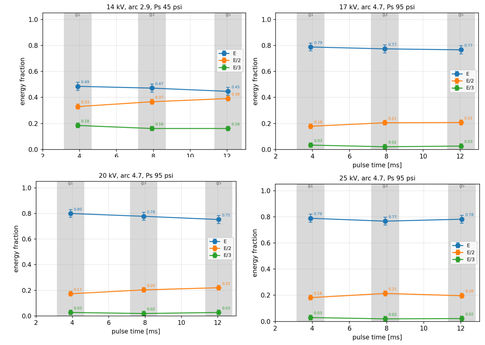


Fig 14. Average species fractions over three time gates across the duration of a shot. In the most off-nominal case, full energy species fraction drops significantly throughout the shot, while in the optical case it stays relatively constant. 3σ uncertainty bars plotted (from empirical repeated shots).

Species fractions vs voltage & perveance

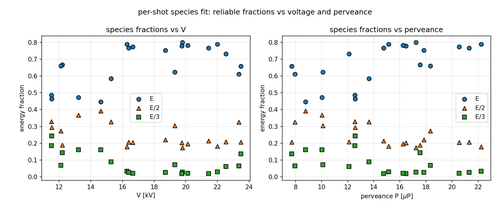


Fig 15. Species fractions as a function of voltage & perveance on WHAM NBI. Optimal operational regime is clearly observed being at higher V & Π due to the highest full energy species fraction.

References

1. A. J. Deka *et al.*, J. Appl. Phys. **123**, 043307 (2018).
2. R. Uhlemann *et al.*, Rev. Sci. Instrum. **64**, 974 (1993).